

1. Patients Table

```
CREATE TABLE Patients (  
    PatientID INT PRIMARY KEY,  
    PatientName VARCHAR(100),  
    Gender CHAR(1),  
    DOB DATE,  
    Phone VARCHAR(15),  
    Address VARCHAR(200)  
);  
  
INSERT INTO Patients VALUES  
(1, 'John Smith', 'M', '1985-06-15', '1234567890', '123 Main St'),  
(2, 'Jane Doe', 'F', '1990-09-25', '2345678901', '456 Oak St'),  
(3, 'Emily Davis', 'F', '1975-03-12', '3456789012', '789 Pine St'),  
(4, 'Michael Brown', 'M', '2000-01-01', '4567890123', '101 Maple St'),  
(5, 'Sarah Lee', 'F', '1988-11-30', '5678901234', '202 Cedar St');
```

2. Doctors Table

```
CREATE TABLE Doctors (  
    DoctorID INT PRIMARY KEY,  
    DoctorName VARCHAR(100),  
    Specialty VARCHAR(50),  
    Phone VARCHAR(15)  
);  
  
INSERT INTO Doctors VALUES  
(1, 'Dr. Wilson', 'Cardiology', '9876543210'),  
(2, 'Dr. Thomas', 'Neurology', '8765432109'),  
(3, 'Dr. Green', 'Orthopedics', '7654321098');
```

3. Appointments Table

```
CREATE TABLE Appointments (  
    AppointmentID INT PRIMARY KEY,  
    PatientID INT FOREIGN KEY REFERENCES Patients(PatientID),  
    DoctorID INT FOREIGN KEY REFERENCES Doctors(DoctorID),  
    AppointmentDate DATE,  
    Status VARCHAR(20)  
);
```

```
INSERT INTO Appointments VALUES  
(1, 1, 1, '2023-07-01', 'Completed'),  
(2, 2, 2, '2023-07-02', 'Completed'),  
(3, 3, 3, '2023-07-03', 'Cancelled'),  
(4, 4, 1, '2023-07-04', 'Completed'),  
(5, 5, 3, '2023-07-05', 'Scheduled'),  
(6, 1, 2, '2023-07-15', 'Completed'),  
(7, 2, 3, '2023-07-18', 'Completed'),  
(8, 3, 1, '2023-07-22', 'Completed');
```

4. Treatments Table

```
CREATE TABLE Treatments (  
    TreatmentID INT PRIMARY KEY,  
    AppointmentID INT FOREIGN KEY REFERENCES  
Appointments(AppointmentID),  
    Diagnosis VARCHAR(200),  
    TreatmentDetails VARCHAR(200),  
    Cost DECIMAL(10, 2)  
);
```

```
INSERT INTO Treatments VALUES  
(1, 1, 'Heart Pain', 'ECG, Aspirin, Rest', 2500.00),  
(2, 2, 'Migraine', 'MRI, Painkillers', 1800.00),  
(3, 4, 'Chest Pain', 'X-ray, Rest', 2000.00),  
(4, 6, 'Headache', 'CT Scan, Paracetamol', 1600.00),  
(5, 7, 'Back Pain', 'Physiotherapy', 2200.00),  
(6, 8, 'Hypertension', 'BP Check, Medication', 1700.00);
```

5. Medications Table

```
CREATE TABLE Medications (  
    MedicationID INT PRIMARY KEY,  
    TreatmentID INT FOREIGN KEY REFERENCES Treatments(TreatmentID),  
    MedicationName VARCHAR(100),  
    Dosage VARCHAR(50)  
);
```

```
INSERT INTO Medications VALUES  
(1, 1, 'Aspirin', '100mg'),  
(2, 1, 'Atorvastatin', '20mg'),  
(3, 2, 'Ibuprofen', '400mg'),  
(4, 3, 'Paracetamol', '500mg'),  
(5, 4, 'Sumatriptan', '50mg'),  
(6, 5, 'Diclofenac', '75mg'),  
(7, 6, 'Amlodipine', '5mg');
```

Easy-Level Questions (Each 1 Marks)

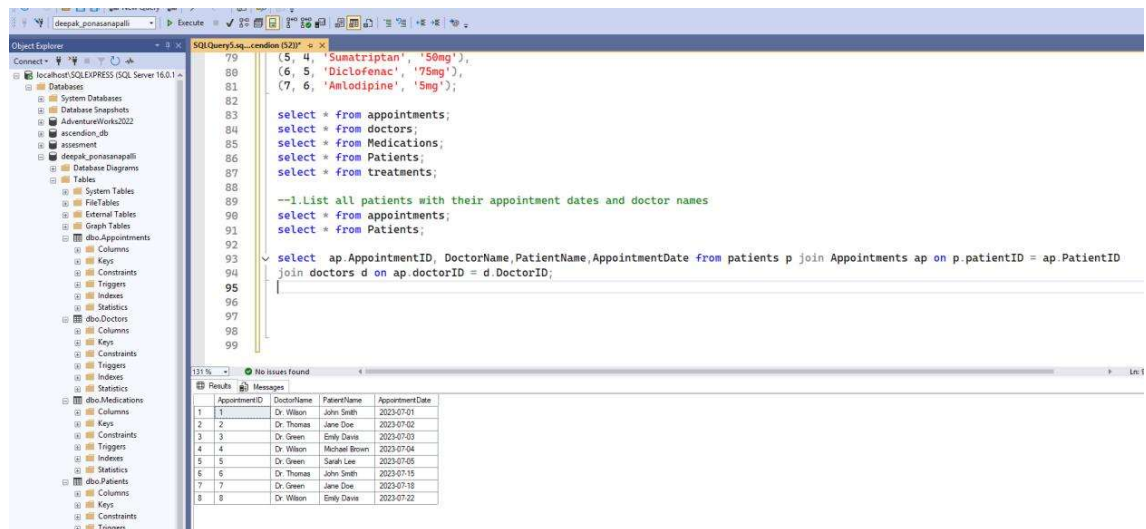
1. List all patients with their appointment dates and doctor names.

```
select * from appointments;
```

```
select * from Patients;
```

```
select * from Doctors
```

```
select ap.AppointmentID, DoctorName, PatientName, AppointmentDate from  
patients p join Appointments ap on p.patientID = ap.PatientID join doctors d on  
ap.doctorID = d.DoctorID;
```



2. Find the total number of appointments per doctor.

```

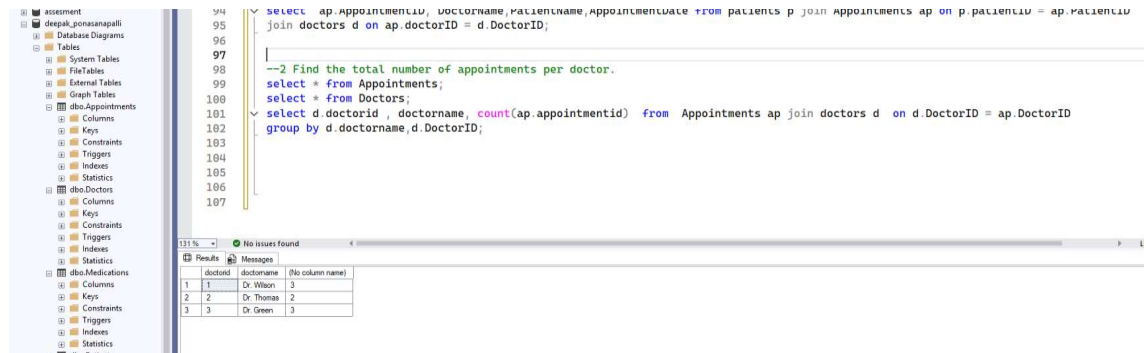
select * from Appointments;
select * from Doctors;

```

```

select d.doctorid , doctorname, count(ap.appointmentid) from Appointments
ap join doctors d on d.DoctorID = ap.DoctorID
group by d.doctorname,d.DoctorID;

```



3. Display the list of medications prescribed during treatment ID = 1.

```

select * from Medications;

```

```

select * from Medications where treatmentid = 1;

```

4. **Get the list of patients who have appointments marked as 'Completed'.**

```
select * from appointments;  
select * from patients;
```

```
select * from doctors;
```

```
select patientName, p.patientid, gender from patients p join appointments ap on  
p.patientid = ap.patientid where Status = 'completed';
```

5. **Show the names and phone numbers of doctors who specialize in 'Neurology'.**

```
select * from doctors;
```

```
select doctorname, phone from doctors where specialty = 'neurology';
```

Medium-Level Questions (Each 3 Marks)

1. **Subquery #1:**

List patients whose total treatment cost is above the average treatment cost of all patients.

```
select * from appointments;  
select * from doctors;  
select * from patients;  
select * from treatments;
```

```
select patientName, sum(cost) from patients p join appointments ap on p.patientid =  
ap.patientid join treatments t on ap.appointmentID = t.appointmentid group by  
patientName having sum(cost) > (select avg(cost) from treatments );
```

2. **Subquery #2:**

List doctors who treated patients with diagnoses containing the word 'Pain'.

```
select * from appointments;  
select * from doctors;  
select * from treatments;
```

```
select doctorname,doctorid from doctors where doctorid in (select ap.DoctorID from
appointments ap join treatments t on ap.appointmentID = t.appointmentID where
t.diagnosis like '%pain%' );
```

3. Window Function #1:

Rank patients by total treatment cost (highest to lowest).

```
select * from appointments;
select * from Patients;
select * from treatments
```

```
select p.patientname, sum(t.cost) as tot, rank() over (order by sum(t.cost) desc) as
cost from appointments ap join patients p on p.PatientID = ap.PatientID join
treatments t on ap.appointmentid = t.appointmentid group by p.PatientName order by
tot desc;
```

4. Window Function #2:

For each doctor, show their appointments ranked by most recent first.

```
select * from appointments;
select * from doctors;
```

```
select doctorname, appointmentdate, appointmentID,rank() over (partition by
ap.doctorid order by appointmentdate desc) as recent
from doctors d join appointments ap on ap.doctorid=d.doctorid
```

5. Date Handling:

Show appointments with extracted month and year of appointment.

```
select * from Appointments;
select * from Patients;
select * from Doctors;
```

```
select appointmentid,doctorname,
month(ap.AppointmentDate) ,year(ap.AppointmentDate) from appointments ap
join patients p on ap.patientID = p.patientID join doctors d on ap.doctorid =
d.doctorid;
```

6. List the total cost of treatments done by each doctor.

```
select * from Appointments;
select * from Doctors;
```

```
select * from Treatments;
```

```
select doctorname, sum(cost) from doctors d join appointments a on d.doctorID =  
a.doctorID join treatments t on a.appointmentid = t.appointmentid group by  
d.doctorname;
```

7. Display patients who have more than 1 appointment.

```
select * from Patients;  
select * from Appointments;
```

```
select patientname,gender,PatientID from Patients where PatientID in ( select  
PatientID from appointments group by patientid having count(appointmentid)>1);
```

8. Find the average treatment cost for each diagnosis.

```
select * from Treatments;  
select Diagnosis, avg(Cost) from treatments group by diagnosis;
```

9. List patients who have not received any treatments yet.

```
select * from Patients;  
select * from Appointments;
```

```
select patientname,p.PatientID from patients p where p.patientid not in (select  
ap.PatientID from appointments ap join treatments t on ap.appointmentid =  
t.appointmentid);
```

10. List all patient and doctor pairs with appointments scheduled for July 2023.

```
select * from Patients;
```

```
select * from Doctors;
```

```
select * from Appointments;
```

```
select patientname, doctorname, ap.appointmentdate from appointments ap join  
patients p ON ap.patientid = p.patientid join doctors d on ap.doctorid = d.doctorid  
where month(ap.appointmentdate) = 7 and year(ap.appointmentdate) = 2023;
```